

Datadog Observability Analysis Report

Dataset: input | **Generated:** 2026-07-10T10:45:28Z **Analysis Period:** 2026-07-02T08:00:00.000Z -> 2026-07-02T10:00:01.000Z **Overall Health:** CRITICAL

Executive Summary

- Total incidents identified: 2
- Critical issues: 25 | Error issues: 19 | Warnings: 10
- Top risks:
 - checkout-consumer identified as origin; failure propagated to order-service, payment-service within 5 minutes
 - Unauthorised access attempt: 403 Forbidden for endpoint /admin/users
 - PII field 'email' detected in log message: user record email=[REDACTED]

1. Errors & Data Quality

Service	Issue	Severity	Detail
checkout-consumer	UNKNOWN	● CRITICAL	Kafka consumer lag critical for topic ecommerce-events, lag=125000
checkout-consumer	UNKNOWN	● CRITICAL	Kafka consumer lag critical for topic ecommerce-events, lag=118000
user-service	UNKNOWN	● CRITICAL	PII field 'email' detected in log message: user record email=[REDACTED]
user-service	UNKNOWN	● CRITICAL	Credential leak detected: Authorization: [REDACTED]
payment-service	CONNECTION_FAILURE	● ERROR	DB connection refused: connection to payments-db timed out

...14 more error(s) not shown; see log_analysis.json.all_errors for the full list.

Worst offending DQ columns:

Column	Rejection Count	Rule Type
email	45	FORMAT_MISMATCH
phone	30	NULL_VALUE

2. Performance & Infrastructure

Latency by service:

Service	Avg (ms)	P95 (ms)	P99 (ms)	Verdict
order-service	2093.8	4140.0	4188.0	CRITICAL
payment-service	1982.5	3855.0	3891.0	CRITICAL
checkout-consumer	3400.0	3490.0	3498.0	CRITICAL
user-service	75.0	88.5	89.7	OK

Slowest traces:

Service	Operation	Duration (ms)	Timestamp
order-service	create_order	4200	2026-07-02T09:30:00.000Z
payment-service	process_payment	3900	2026-07-02T09:30:01.000Z
order-service	create_order	3800	2026-07-02T09:31:00.000Z
payment-service	process_payment	3600	2026-07-02T09:31:02.000Z
checkout-consumer	consume_event	3500	2026-07-02T09:29:50.000Z

Host resource health:

Host	CPU %	Memory %	Disk %	Verdict	Issues
host-01	95.0	91.0	43.0	CRITICAL	HIGH_CPU, HIGH_MEMORY, NETWORK_SATURATION, RESOURCE_EXHAUSTION, HOST_DOWN
host-03	88.0	82.0	61.0	CRITICAL	HIGH_CPU, HIGH_MEMORY, NETWORK_SATURATION, HOST_DOWN
host-02	72.0	68.0	36.0	WARN	HOST_DOWN
host-04	42.0	44.0	31.0	OK	-

All performance/infra issues:

Service	Issue	Severity
order-service	HIGH_ERROR_RATE	CRITICAL
payment-service	HIGH_ERROR_RATE	CRITICAL
checkout-consumer	HIGH_ERROR_RATE	CRITICAL
user-service	HIGH_ERROR_RATE	CRITICAL
host-01	HIGH_CPU	CRITICAL

Service	Issue	Severity
host-01	HIGH_MEMORY	 CRITICAL
host-01	NETWORK_SATURATION	 CRITICAL
host-01	RESOURCE_EXHAUSTION	 CRITICAL
host-01	HOST_DOWN	 CRITICAL
host-03	HIGH_CPU	 CRITICAL
host-03	HIGH_MEMORY	 CRITICAL
host-03	NETWORK_SATURATION	 CRITICAL
host-03	HOST_DOWN	 CRITICAL
order-service	HIGH_LATENCY	 CRITICAL
payment-service	HIGH_LATENCY	 CRITICAL
checkout-consumer	HIGH_LATENCY	 CRITICAL

...2 more issue(s) not shown; see `metrics_report.json.all_issues` for the full list.

3. Pipeline Health

Kafka topics:

Topic	Consumer Group	Lag	Verdict
(unknown)	checkout-consumer	8000.0	 OK
(unknown)	checkout-consumer	25000.0	 WARN
(unknown)	checkout-consumer	68000.0	 WARN
ecommerce-events	checkout-consumer	125000.0	 CRITICAL
ecommerce-events	checkout-consumer	118000.0	 CRITICAL

All pipeline issues:

Service	Issue	Severity
checkout-consumer	KAFKA_LAG_CRITICAL	 CRITICAL
checkout-consumer	KAFKA_LAG_CRITICAL	 CRITICAL
checkout-consumer	CHECKPOINT_STALE	 CRITICAL
checkout-consumer	KAFKA_LAG_HIGH	 WARN
checkout-consumer	KAFKA_LAG_HIGH	 WARN

...1 more issue(s) not shown; see `apm_report.json.all_issues` for the full list.

4. Security

Service	Issue	Severity	Detail
user-service	PII_IN_LOGS	CRITICAL	PII field 'email' detected in log message: user record email=[REDACTED]
user-service	CREDENTIAL_LEAK	CRITICAL	Credential leak detected: Authorization: [REDACTED]
user-service	UNAUTHORISED_ACCESS	ERROR	Unauthorised access attempt: 403 Forbidden for endpoint /admin/users
user-service	UNAUTHORISED_ACCESS	ERROR	Unauthorised access attempt: 401 Unauthorized invalid token
user-service	UNAUTHORISED_ACCESS	ERROR	Unauthorised access attempt: 401 Unauthorized invalid token

5. Anomalies & Trends

Service	Anomaly	Confidence	Detail
order-service	CORRELATED_ANOMALY	HIGH	order-service and checkout-consumer showed correlated KAFKA_LAG_SPIKE and LATENCY_SPIKE within the same window
checkout-consumer	KAFKA_LAG_SPIKE	MEDIUM	checkout-consumer kafka_consumer_lag reading 125000.0 exceeded 2.0x its own prior baseline 33666.67
order-service	LATENCY_SPIKE	MEDIUM	order-service p99 latency 4188.0ms is 2.0x its own baseline avg 2093.8ms

6. Dependency & Breakpoint Analysis

Service graph: 3 node(s), 2 edge(s).

Breakpoint Service	Downstream Impact	Confidence
checkout-consumer	order-service, payment-service	0.65

Root Cause Analysis

incident_005 — PIPELINE_BACKPRESSURE (confidence: HIGH, severity: CRITICAL, blast radius: 3)

- Primary service: checkout-consumer
- Root cause: checkout-consumer identified as origin; failure propagated to order-service, payment-service within 5 minutes
- Downstream symptoms: KAFKA_LAG_HIGH on checkout-consumer; DQ_ALERT rejection_reason=FORMAT_MISMATCH:email count=45; DQ_ALERT

rejection_reason=NULL_VALUE;phone count=30; Kafka consumer lag critical for topic ecommerce-events, lag=125000; KAFKA_LAG_CRITICAL on checkout-consumer; order-service and checkout-consumer showed correlated KAFKA_LAG_SPIKE and LATENCY_SPIKE within the same window; checkout-consumer kafka_consumer_lag reading 125000.0 exceeded 2.0x its own prior baseline 33666.67; checkout-consumer error rate 100.0% across 2 traces; checkout-consumer p99 latency 3498.0ms; Request timeout: upstream checkout-consumer not responding; order-service error rate 50.0% across 4 traces; payment-service throughput dropped 46.8% vs baseline; order-service p99 latency 4188.0ms; order-service p99 latency 4188.0ms is 2.0x its own baseline avg 2093.8ms; payment-service error rate 50.0% across 4 traces; payment-service p99 latency 3891.0ms; Unhandled exception: NullPointerException at OrderProcessor.finalize; CHECKPOINT_STALE on checkout-consumer; Kafka consumer lag critical for topic ecommerce-events, lag=118000

- Evidence sources: anomaly_report.json, apm_report.json, dependency_report.json, log_analysis.json, metrics_report.json

incident_006 — SECURITY_INCIDENT (confidence: HIGH, severity: CRITICAL, blast radius: 1)

- Primary service: user-service
- Root cause: Unauthorised access attempt: 403 Forbidden for endpoint /admin/users
- Downstream symptoms: Unauthorised access attempt: 403 Forbidden for endpoint /admin/users; Unauthorised access attempt: 401 Unauthorized invalid token; user-service error rate 50.0% across 2 traces; External monitor 'user-service-auth-failures' reported possible brute force -- not independently confirmed; PII field 'email' detected in log message: user record email=[REDACTED]; Credential leak detected: Authorization:[REDACTED]
- Evidence sources: log_analysis.json, metrics_report.json, security_report.json

⚠️ 9 critical finding(s) could not be clustered into an incident -- see [root_cause.json.unresolved_findings](#).

Recommendations

Rank	Priority	Title	Affected Services
1	P1_IMMEDIATE	Address PIPELINE_BACKPRESSURE on checkout-consumer	checkout-consumer, order-service, payment-service
2	P1_IMMEDIATE	Address SECURITY_INCIDENT on user-service	user-service

Patch Suggestions (Human Review Required)

⚠️ All patches require human review before applying.

patch_001 (MEDIUM risk, SCALING_CONFIG_CHANGE) — targets [config/kafka-consumer.yaml](#)

- Observed consumer lag peaked at 125000; scale consumer instances to 4 to relieve backpressure (assumes a starting scale of 2 -- confirm actual current value before applying).

```
- consumer_instances: 2
+ consumer_instances: 4
```

patch_002 (MEDIUM risk, LOGGING_REDACTION_ADD) — targets `src/logging/redaction.py`

- Log line(s) exposed unredacted Authorization, email; add Authorization, email to the redaction field list before logging.

```
- redact_fields = ["password", "token"]
+ redact_fields = ["password", "token", "Authorization", "email"]
```

Appendix — Ingestion Summary

Source Type	Record Count
log	26
metric	35
trace	12
alert	7
infrastructure	10

Total records ingested: 90